PROGRAM-1

AIM:
Write a Java program to accept following details about a student as follows:

i. rollno
ii. fullname
iii. address
iv. Stream
v. total marks in 5 subjects
vi. percentage

display all the details in a readable format?

CODE:

```
package mca1;
import java.util.Scanner;
public class QU1 {

	public static void main(String[] args) {
		Scanner sc = new Scanner(System.in);
		System.out.println("enter roll no");
		int roll = sc.nextInt();
		System.out.println("enter your fullname");
		String fullname= sc.next();
		System.out.println("enter your address");
		String add= sc.next();
		System.out.println("enter your stream");
		String strm = sc.next();
		System.out.println("enter your marks in 1st subject");
		int m1= sc.nextInt();
		System.out.println("enter your marks in 2nd subject");
		int m2= sc.nextInt();
		System.out.println("enter your marks in 3rd subject");
		int m3= sc.nextInt();
		System.out.println("enter your marks in 4th subject");
		int m4= sc.nextInt();
		System.out.println("enter your marks in 5th subject");
		int m5= sc.nextInt();
		int t = m1+m2+m3+m4+m5;
		int per=  t/5;
		System.out.println("roll no:" +roll);
		System.out.println("fullname:" + fullname);
		System.out.println("address:" +add);
		System.out.println("stream:" +strm);
		System.out.println("total marks:" +t);
		System.out.println("percentage:" +per + "%");
		}
```

}

OUTPUT:

```
enter roll no
57
enter your fullname
devjoshi
enter your address
meerut
enter your stream
mca
enter your marks in 1st subject
70
enter your marks in 2nd subject
75
enter your marks in 3rd subject
80
enter your marks in 4th subject
75
enter your marks in 5th subject
70
roll no:57
fullname:devjoshi
address:meerut
stream:mca
total marks:370
percentage:74%
```

Program-2

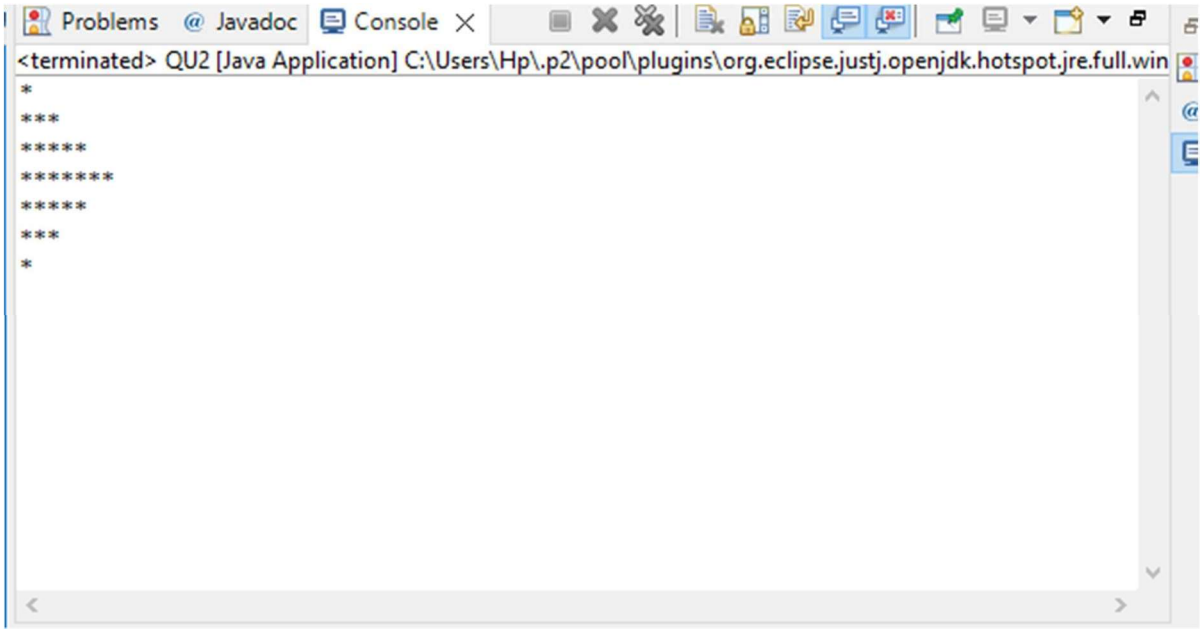
AIM: Write a java program to print following output:

```
*
***
*****
*******
*****
***
*
```

CODE:

```
package mca1;

public class QU2 {

	public static void main(String[] args) {
		for(int i=1; i<=4; i++) {
			for(int j=1; j<= (2*i)-1; j++ ) {
				System.out.print("*");
			}
			System.out.println();

		}
		for(int i=1; i<=3;i++)
		{
			for (int j=5; j>=2*i -1 ; j--) {
				System.out.print("*");
				}
			System.out.println();
}
		}
	}
```

Output:


Problems
Javadoc
Console
<terminated> QU2 [Java Application] C:\Users\Hp\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win
*
***
*****
****
*****
***
*

Program-3

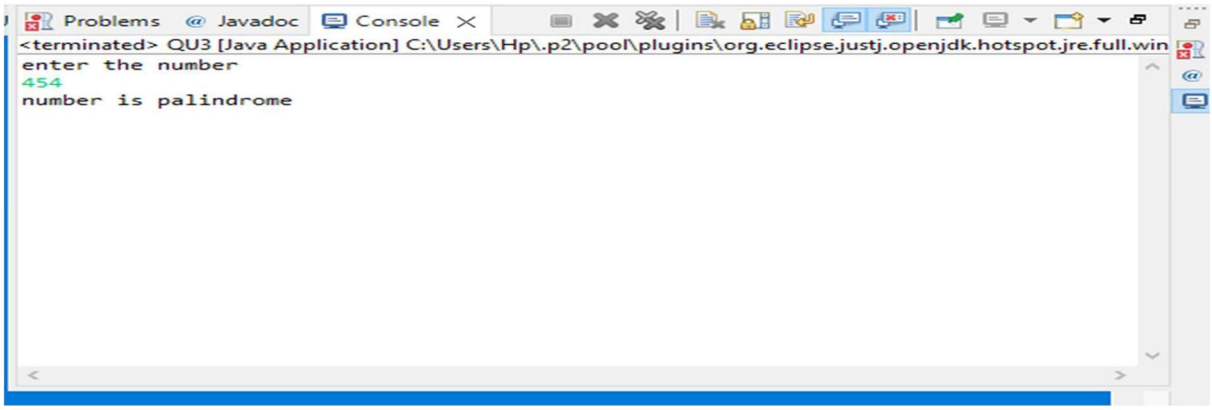
AIM:

Write a java program to check if a number entered by the user is “palindrome” or not.

Code:

```
package mca1;

import java.util.Scanner;

public class QU3 {

        public static void main(String[] args) {
                Scanner sc= new Scanner (System.in);
                System.out.println("enter the number");
                int no = sc.nextInt();
      int num = no;
int rev=0,rem;
                while(num>0) {
                rem = num%10;
                rev= (rev *10)+ rem;
                num= num/10;
                }
                if(no==rev)
                {
                        System.out.println("number is palindrome");
                }
                        else
                        {
                                System.out.println("number is not palindrome");
                        }
                }
        }
```

Output:



Program-4

Aim:
Write a java program to print tables from 0 to accepted numbers, using loops and keyboard inputs.

Code:

```
package mca1;
import java.util.Scanner;

public class QU4 {

        public static void main(String[] args) {
                Scanner sc=new Scanner(System.in);
            System.out.println("Enter the number");
            int n=sc.nextInt();
            System.out.println("Table from 1 to "+n+":-");
            for(int i=0;i<=n; i++){
              for(int j=1;j<=10;j++){
                System.out.print(i*j+" ");
              }
              System.out.println();
              sc.close();
              }
}
}
```

Output:

```
Problems  @ Javadoc  Console
<terminated> QU4 [Java Application] C:\Users\Hp\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspc
Enter the number
6
Table from 1 to 6:-
0 0 0 0 0 0 0 0 0 0
1 2 3 4 5 6 7 8 9 10
2 4 6 8 10 12 14 16 18 20
3 6 9 12 15 18 21 24 27 30
4 8 12 16 20 24 28 32 36 40
5 10 15 20 25 30 35 40 45 50
6 12 18 24 30 36 42 48 54 60
```

Program-5

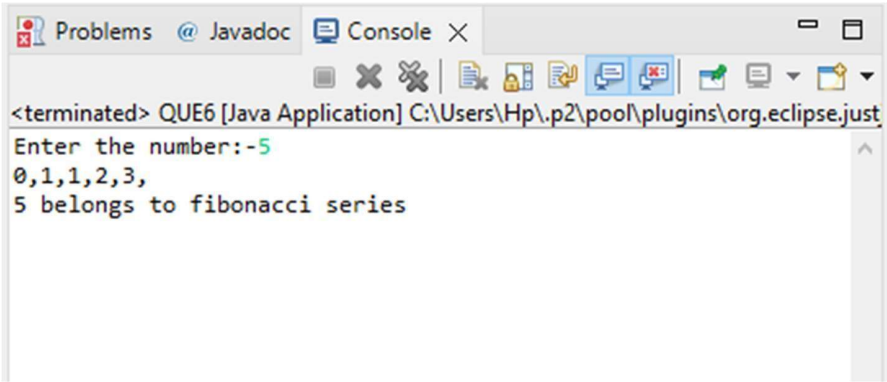
Aim:

Write a java program to check input no is part of Fibonacci series or not? Print Fibonacci series till that point.

Code:

```
package mca1;
import java.util.Scanner;
public class QUE6 {
        public static void main(String[ ] args)
        {
        Scanner sc=new Scanner (System.in);
        System.out.print ("Enter the number:-");
        int n=sc.nextInt();
        int  n1=0,n2=1,n3=0;
        while (n3<=n) {
        System.out.print (n1+ ",");
        n3=n1+n2;
        n1=n2;
        n2=n3;
        }
        System.out.println();
        if(n1==n){
        System.out.println (+n +" " + "belongs to fibonacci series");
        }
        else
        {
        System.out.println (+n + " does not belongs to Fibonacci series");
        }
        }
}
```

Output:



Program-6

Aim :
WAP to remove duplicate elements from the array using a temporary array.

Code:

```
import java.util.*;
public class Q6
{
   public static void main(String[] args)
   {
      Scanner sc = new Scanner(System.in);
      System.out.println("Enter array size");
      int n=sc.nextInt();
      int[] org = new int[n];
       System.out.print("Enter your array elements: ");
      for (int i = 0; i < n; i++) {
         org[i] = sc.nextInt();
      }
      int[] res = remove(org);
      System.out.println(Arrays.toString(res));
      sc.close();
   }
   public static int[] remove(int[] arr)
   {
      List<Integer> tempList = new ArrayList<>();

      for (int element : arr)
      {
         if (!tempList.contains(element))
         {
            tempList.add(element);
         }
      }
      int[] resultArray = new int[tempList.size()];
      System.out.print("Your array after removing duplicacy:- ");
      for (int i = 0; i < tempList.size(); i++)
      {

         resultArray[i] = tempList.get(i);
      }
      return resultArray;
   }
}
```

Output:

```
Problems   @ Javadoc   Console
<terminated> QU6 [Java Application] C:\Users\Hp\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_17.0.8.v20230831-104
Enter array size
6
Enter your array elements: 1 3 5 4 6 7
Your array after removing duplicacy:- [1, 3, 5, 4, 6, 7]
```

Program-7

Aim:

Write a java program to accept 10 integer values from the user, store them in an array,

vii. arrange the array in ascending and descending order,

viii. find the Maximum, minimum and average.

ix. Print only either Odd or Even

Code:

```
package mca1;
import java.util.Scanner;
import java.util.Arrays;

public class QU7{
public static void main(String[] args) {

int n=10;
Scanner sc=new Scanner(System.in);
 int a[]=new int[n];
 System.out.print("Enter 10 elements of array: ");
 for(int i=0;i<n;i++)
a[i]=sc.nextInt();
 System.out.print("Entered elements of array: ");
 for (int num : a) {
 System.out.print(num + " ");
 }
 System.out.println("");
 Arrays.sort(a);
 System.out.print("Array in ascending order: ");
 for (int num : a) {
 System.out.print(num + " ");
 }
 for(int i=0;i<n/2;i++) {
 int temp=a[i];
 a[i]=a[n-i-1];
 a[n-i-1]=temp;
 }
 System.out.println("");
 System.out.print("Array in descending order: ");
 for (int num : a) {
 System.out.print(num + " ");
 }
 int sum=0;
 float avg=0;
 for (int num : a) {
```

```
sum=sum+num;
}
avg = (float)sum/10;
System.out.print("\nMaximum = "+a[0]+"\nMinimum = " +a[n-1] + "\nAverag is : ");
System.out.print(avg);
System.out.print("\nEven elements: ");
for(int num : a) {
if(num%2==0) {
System.out.print(num+" ");
}
}
System.out.print("\nOdd elements: ");
for(int num : a) {
if(num%2==1) {
System.out.print(num+" ");
}
}
}
}
```

Output:

```
Problems  @ Javadoc  Console ×
<terminated> QU7 [Java Application] C:\Users\Hp\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_17.0.8.v20230831
Enter 10 elements of array:
1 2 3 5 7 8 9 10 66 21
Entered elements of array: 1 2 3 5 7 8 9 10 66 21
Array in ascending order: 1 2 3 5 7 8 9 10 21 66
Array in descending order: 66 21 10 9 8 7 5 3 2 1
Maximum = 66
Minimum = 1
Averag is : 13.2
Even elements: 66 10 8 2
Odd elements: 21 9 7 5 3 1
```

Program-8

Aim:
Write a java program to create a calculator. Use classes and methods to perform +,-,*,/,%.

Code:

```
package mca1;
import java.util.Scanner;
class cal{
   public static int add(int a,int b){
   return a+b;
   }
   public static int sub(int a,int b){
   return a-b;
   }
    public static int mul(int a,int b){
   return a*b;
   }
    public static int div(int a,int b){
   return a/b;
   }
    public static int mod(int a,int b){
    if (b == 0) {
          System.out.println("Cannot calculate modulo with zero.");
       }
       return a % b;
   }
}
public class QU8 {
   public static void main(String[] args) {
       Scanner sc=new Scanner(System.in);
       System.out.println("Select an Operation:");
       System.out.println("1. Addition (+)");
       System.out.println("2. Subtraction (-)");
       System.out.println("3. Multiplication (*)");
       System.out.println("4. Division (/)");
       System.out.println("5. Modulo (%)");
       System.out.print("Enter  your  choice:-");
       int n=sc.nextInt();
          System.out.print("Enter 1st Number: ");
       int   a=sc.nextInt();
       System.out.print("Enter 2nd number: ");
       int b=sc.nextInt();

       switch (n) {
          case 1:
```

```
            System.out.println("Result: "+cal.add(a,b));
            break;
        case 2:
            System.out.println("Result: "+cal.sub(a,b));
            break;
        case 3:
            System.out.println("Result: "+cal.mul(a,b));
            break;
        case 4:
            System.out.println("Result: "+cal.div(a,b));
            break;
        case 5:
            System.out.println("Result: "+cal.mod(a,b));
            break;
        default:
        System.out.println("Invalid selection");
            break;
    }
    sc.close();
  }
}
```

Output:

```
Problems  @ Javadoc  Console ×
<terminated> QU8 [Java Application] C:\Users\Hp\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_17.0.8.v20230831-1047\jre\bin\javaw.exe
Select an Operation:
1. Addition (+)
2. Subtraction (-)
3. Multiplication (*)
4. Division (/)
5. Modulo (%)
Enter your choice:-4
Enter 1st Number: 12
Enter 2nd number: 6
Result: 2
```

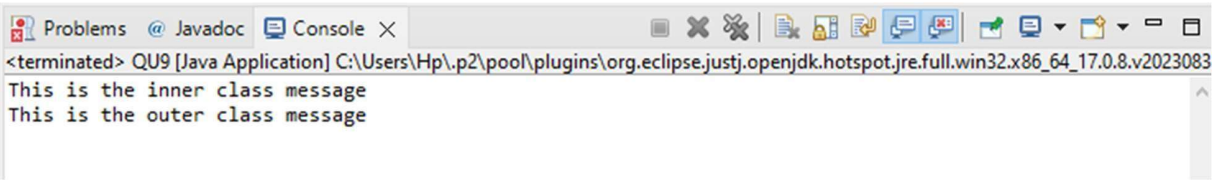
Program-9

Aim:
Create a class “Enclosed” within it create inner class “Nested”, both the classes should have at least one method to display messages. Try to call the method of the “Nested” class in the “Enclosed” class and vice versa.

Code:

```
package mca1;
public class QU9 {
   private String outerMessage = "This is the outer class message";
   public void displayOuterMessage() {
      System.out.println(outerMessage);
   }
   public class Nested {
      private String innerMessage = "This is the inner class message";
      public void displayInnerMessage() {
         System.out.println(innerMessage);
      }
      public void callOuterClassMethod() {
         displayOuterMessage();
      }
   }
   public void callInnerClassMethod() {
      Nested nested = new Nested();
      nested.displayInnerMessage();
   }
   public static void main(String[] args) {
      QU9 outer = new QU9();
      QU9.Nested inner = outer.new Nested();
      outer.callInnerClassMethod();
      inner.callOuterClassMethod();
   }
}
```

Output:



Program-10

Aim:

Create a class called MyString : Declare two string type variables: str1 (“ Welcome to Java tutorial”) and str2(“Today's topic is String Handling in Java”). Perform following operations in this class:

x. Concatenate two strings
xi. Covert str1 into lower case
xii. Covert str2 into upper case
xiii. Are both equal to each other
xiv. Show the location of “J” in both str1 and str2
xv. Replace “i” with “I” in both the strings
xvi. display “java” from str string
xvii. Display the “7” character in str1.
xviii. Convert str1 into string array

Code:

```
package mca1;
import java.lang.*;

public class QU10 {

        public static void  main(String[]  args)  {
                String str="Welcome to Java tutorial";
                String str2="Today's topic is String Handling in Java";
                System.out.println("Concatinating two string : "+str.concat(str2));
                System.out.println("toLowerCase(): "+str.toLowerCase());
                System.out.println("toUpperCase(): "+str2.toUpperCase());
                System.out.println("String is equal? : "+str.equals(str2));
                System.out.printf("'J' is at index %d in str.", str.indexOf( 'J' ));
                System.out.printf("\n'J' is at index %d in str2.", str2.indexOf( 'J' ));
                System.out.printf("\nstring after replacing i with I in str: "+str.replace('i','I'));
                System.out.printf("\nstring after replacing i with I in str2:
"+str2.replace('i','I')+"\n");
                String s="java";
                int n=str.indexOf(s);
                if(n>=0){
                System.out.println(s);
                }
                else{
                System.out.println("java is not found in str");
                }
                System.out.println("Seventh character is "+str.charAt(6));
                String[] arr = new String[] {str2};
```

```
System.out.printf("string print in using array: ");
for(String ch : arr)
{
System.out.println(ch);
}
}

}
```

Output:

```
Problems  @ Javadoc  Console
<terminated> QU10 [Java Application] C:\Users\Hp\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_17.0.8.v2
Concatinating two string : Welcome to Java tutorialToday's topic is String Handling in Java
toLowerCase(): welcome to java tutorial
toUpperCase(): TODAY'S TOPIC IS STRING HANDLING IN JAVA
String is equal? : false
'J' is at index 11 in str.
'J' is at index 36 in str2.
string after replacing i with I in str: Welcome to Java tutorIal
string after replacing i with I in str2: Today's topIc Is StrIng HandlIng In Java
java is not found in str
Seventh character is e
string print in using array: Today's topic is String Handling in Java
```

Program-11

Aim:
Create a class person (Data Member: Name & address , Method: Accept() and display() to accept and display value of data member on Output device. Derive two classes student (Data Member: Rollno, Course Member Method: Accept() and Display()) and Employee((Data Member: EmpId ,Department Member Method: Accept() and Display()).display details of one student and one employee. NOTE: use super keyword to invoke hidden members of base class.

code:

```
package mca1;


class person{
void accept() {
	String name;
	String address;
}
void display() {
	System.out.println("name : dev");
}
}

class students extends person{
	void accept() {
		int rollno;
		String course;
	}
	void display() {
		super.display();
	 System.out.println("rollno:  57");
		System.out.println("course:  mca");
		}
}

class employee extends students{
	void accept() {
		int EmpID;
		String departmentmemeber;
		}
	void display() {
		super.display();
		System.out.println("empid:  23");
		System.out.println("departmentmember:  yes");
	}
}
public class QU1 {
 public static void main(String[] args) {
	 employee obj = new employee();
	 obj.display();
	}
```

```
}
```

Output:

```
name : dev
rollno: 57
course: mca
empid: 23
departmentmember: yes
```

Program-12

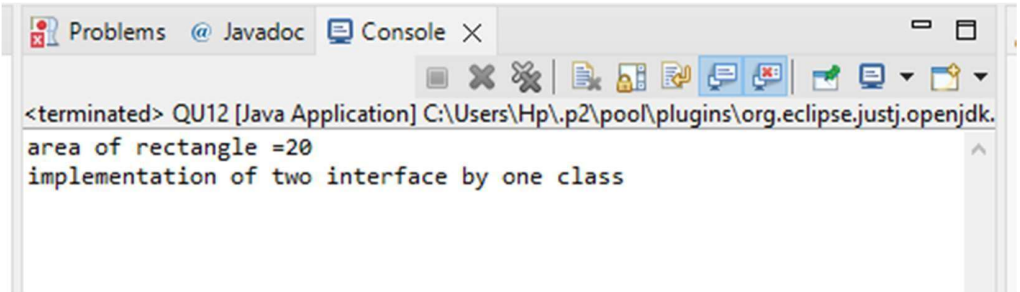
Aim:

WAP to show the use of Interfaces in java.

Code:

```
package mca1;

//create interface cal
interface cal{
        public void getArea(int length, int breadth);
}
//create second interface answer
interface answer {
        void display(); //interface doesnt have method body
}
//implements interface in class
//rectangle class implements 2 interfaces
 class Rectangle implements cal, answer{
        public void getArea(int length, int breadth) {
                 System.out.println("area of rectangle " + "=" +(length * breadth));
         }
        public void display(){
                System.out.println("implementation of two interface by one class");
        }
         }
 //call method
public class QU12 {
public static void main(String[] args) {
          Rectangle obj = new Rectangle();
                obj.getArea(4,5);
                obj.display();
    }
}
```

Output:



Program-13

Aim:
Write a java program to display the grade of students depending on marks, please raise a user defined checked Exception, if less than 0 or more than 100 marks are entered for grade.

Code:

```
package mca1;
import java.util.Scanner;
class std extends Exception {
   public std(String message) {
      super(message);
   }
}
class GradeCalculator {
   public char calculateGrade(int marks) throws std {
      if (marks < 0 || marks > 100) {
         throw new std("Marks should be between 0 and 100.");
      }
      if (marks >= 90) {
         return 'A';
      } else if (marks >= 80) {
         return 'B';
      } else if (marks >= 70) {
         return 'C';
      } else if (marks >= 60) {
         return 'D';
      } else {
         return 'F';
      }
   }
}
public class QU20 {
   public static void main(String[] args) {
      try {
         Scanner sc=new Scanner(System.in);
         System.out.println("Enter your marks:-");
         int studentMarks = sc.nextInt();
         GradeCalculator calculator = new GradeCalculator();
         char grade = calculator.calculateGrade(studentMarks);
         System.out.println("Grade: " + grade);
         sc.close();
      } catch (std e) {
         System.err.println("Error: " + e.getMessage());
      }
   }
```

}

Output:

```
Problems  @ Javadoc  Console
<terminated> QU20 [Java Application] C:\Users\Hp\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_17.0.8.v20230831
Enter your marks:-
120
Error: Marks should be between 0 and 100.
```

Program-14

Aim:
Write a Java program to use the try and catch and finally block.

Code:

```
package mca1;

public class QU14 {

        public static void main(String[] args) {
                try {
                    int a = 15;
                    int b = 0;

                    System.out.println("Value of a = " + a);
                    System.out.println("Value of b = " + b);

                    int c = a / b;
                    System.out.println("a / b = " + c);
                  }
                  catch (Exception e) {
                    System.out.println("Exception Thrown: " + e);
                  }
                  finally {
                    System.out.println("Finally block executed!");
                  }
                }
               }
```

Output:

```
Problems  @ Javadoc  Console ×
<terminated> QU14 [Java Application] C:\Users\Hp\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_17.0.8.v20230831-1047\jre\bin\javaw.exe (1
Value of  a = 15
Value of  b = 0
Exception Thrown: java.lang.ArithmeticException: / by zero
Finally block executed!
```

Program-15

Aim:
Write a multithreaded program where one thread will print 0-5 and another thread will print 5-0. Use thread class.

Code:

```
package mca1;

class NumberPrinter extends Thread {
   private int start;
   private int end;
   public NumberPrinter(int start, int end) {
      this.start = start;
      this.end = end;
   }

 @Override
   public void run() {
      if (start < end) {
         for (int i = start; i <= end; i++) {
            System.out.println(Thread.currentThread().getName() + ": " + i);
         }
      } else {
         for (int i = start; i >= end; i--) {
            System.out.println(Thread.currentThread().getName() + ": " + i);
         }
      }
   }
}

public class QUE15 {
   public static void main(String[] args) {
      Thread thread1 = new NumberPrinter(0, 5);
      Thread thread2 = new NumberPrinter(5, 0);
      thread1.start();
      thread2.start();
   }
}
```

Output:

```
Problems  @ Javadoc  Console ×
<terminated> QUE15 [Java Application] C:\Users\Hp\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_17.0.8.v20230831-1047\jre\bin\javaw.exe (
Thread-1: 5
Thread-0: 0
Thread-0: 1
Thread-0: 2
Thread-0: 3
Thread-0: 4
Thread-0: 5
Thread-1: 4
Thread-1: 3
Thread-1: 2
Thread-1: 1
Thread-1: 0
```

Program-16

Aim:

Write a java multithreaded (2 or more)java program , one thread will print odd numbers and another will print even numbers and the main thread is there it will print date and time. Use Runnable interface

Code:

```
import java.text.SimpleDateFormat;
import java.util.Date;
class Even implements Runnable {
   public void run() {
      for (int i = 2; i <= 20; i += 2) {
         System.out.println("Even: " + i);
         try {
            Thread.sleep(500);
         } catch (InterruptedException e) {
            e.printStackTrace();
         }
      }
   }
}
class Odd implements Runnable {
   public void run() {
      for (int i = 1; i <= 19; i += 2) {
         System.out.println("Odd: " + i);
         try {
            Thread.sleep(500);
         } catch (InterruptedException e) {
            e.printStackTrace();
         }
      }
   }
}
public class Q16 {
   public static void main(String[] args) {
      Thread evenThread = new Thread(new Even());
      Thread oddThread = new Thread(new Odd());
      evenThread.start();
      oddThread.start();
         SimpleDateFormat sdf = new SimpleDateFormat("dd-MM-yyyy HH:mm:ss");
         String dateTime = sdf.format(new Date());
         System.out.println("Date and Time: " + dateTime);
   }
}
```

```java
import java.text.SimpleDateFormat;
import java.util.Date;
class Even implements Runnable {
    public void run() {
        for (int i = 2; i <= 20; i += 2) {
            System.out.println("Even: " + i);
            try {
                Thread.sleep(500);
            } catch (InterruptedException e) {
                e.printStackTrace();
            }
        }
    }
}
class Odd implements Runnable {
    public void run() {
        for (int i = 1; i <= 19; i += 2) {
            System.out.println("Odd: " + i);
            try {
                Thread.sleep(500);
            } catch (InterruptedException e) {
                e.printStackTrace();
            }
        }
    }
}
public class QU16 {
    public static void main(String[] args) {
        Thread evenThread = new Thread(new Even());
        Thread oddThread = new Thread(new Odd());
        evenThread.start();
        oddThread.start();
            SimpleDateFormat sdf = new SimpleDateFormat("dd-MM-yyyy HH:mm:ss");
            String dateTime = sdf.format(new Date());
            System.out.println("Date and Time: " + dateTime);
    }
}
```

Output:

```
Problems  @ Javadoc  Console ×
<terminated> QU16 [Java Application] C:\Users\Hp\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_17.0.8.v20230831-1047\jre\b
Even: 2
Odd: 1
Odd: 3
Even: 4
Even: 6
Odd: 5
Date and Time: 12-10-2023 00:54:12
Even: 8
Odd: 7
Even: 10
Odd: 9
Even: 12
Odd: 11
Odd: 13
Even: 14
Odd: 15
Even: 16
Odd: 17
Even: 18
Odd: 19
Even: 20
```

Program-17

Aim:
WAP to show the use of Legacy classes:- Vector

Code:

```
package mca1;
import java.util.*;
public class QU17 {
   public static void main(String[] args) {
      Vector<String> vector = new Vector<>();
      vector.add("up");
      vector.add("bihar");
      vector.add("bharat");
      Enumeration<String> enumeration = vector.elements();
      while (enumeration.hasMoreElements()) {
         String element = enumeration.nextElement();
         System.out.println("Element: " + element);
      }
      vector.add("state");
      vector.add("country");
      for (String element : vector) {
         System.out.println("Element: " + element);
      }
   }
}
```

Output:

```
Problems  @ Javadoc  Console ×
<terminated> QU17 [Java Application] C:\Users\Hp\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_17.0.8.v20230831-1047\jre\bin\javaw.exe (12-Oc
Element: up
Element: bihar
Element: bharat
Element: up
Element: bihar
Element: bharat
Element: state
Element: country
```

Program-18

Aim:
WAP to show the use of Legacy classes:- Stack

Code:

```
package mca1;
import java.util.*;
public class QU18 {
   public static void main(String[] args) {
      Stack<String> stack = new Stack<>();
      stack.push("x");
      stack.push("y");
      stack.push("z");
      String poppedElement = stack.pop();
      System.out.println("Popped: " + poppedElement);
      String topElement = stack.peek();
      System.out.println("Top element: " + topElement);
      boolean isEmpty = stack.isEmpty();
      System.out.println("Is the stack empty? " + isEmpty);
      int index = stack.search("y"
                  + "");
      System.out.println("Index of 'y' in the stack: " + index);
   }
}
```

Output:

```
Problems  @ Javadoc  Console ×
<terminated> QU18 [Java Application] C:\Users\Hp\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_17.0.8.v20230831-1047\jre\bin\j
Popped: z
Top element: y
Is the stack empty? false
Index of 'y' in the stack: 1
```

Program-19

Aim:

WAP to demonstrate the use of followings:

xix. StringTokenizer

Code:

```
package mca1;
import java.util.StringTokenizer;
public class QU19 {

	public static void main(String[] args) {
		//stringtokenizer
		StringTokenizer st= new StringTokenizer("learn string tokenizer");
		System.out.println("NO. of token is:" +st.countTokens());
		while(st.hasMoreElements()) {
			System.out.println("Next Token is:" +st.nextElement());
			}
	}
}
```

Output:

```
Problems  @ Javadoc  Console ×
<terminated> QU19 [Java Application] C:\Users\Hp\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_17.0.8.v20230831-1047\jre\
NO. of token is:3
Next Token is:learn
Next Token is:string
Next Token is:tokenizer
```

xx. Date

Code:

```
package mca1;
import java.util.Date;
public class QU19B {

	public static void main(String[] args) {
		Date d1 = new Date(); //instantiate a date object

		//display date using tostring()
```

```
            System.out.println(d1.toString());
        }
            }
```

Output:

```
Problems  @ Javadoc  Console ×
<terminated> QU19B [Java Application] C:\Users\Hp\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_17.0.8.v20230831-1047
Thu Oct 12 00:06:54 IST 2023
```

xxi. Calendar

Code:

```
package mca1;
import java.util.*;
import java.util.Calendar;

public class QU19C {

        public static void main(String[] args) {
            //creating an object
            Calendar c = Calendar.getInstance();

            //to get the current date n time
            System.out.print("current date and time is:" + c.getTime());
        }
        }
```

Output:

```
Problems  @ Javadoc  Console ×
<terminated> QU19C [Java Application] C:\Users\Hp\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_17.0.8.v20230831-1047\j
current date and time is:Thu Oct 12 00:07:58 IST 2023
```

Program-20

Aim:

WAP to create a Simple GUI with text field button and label and handle click event of button.

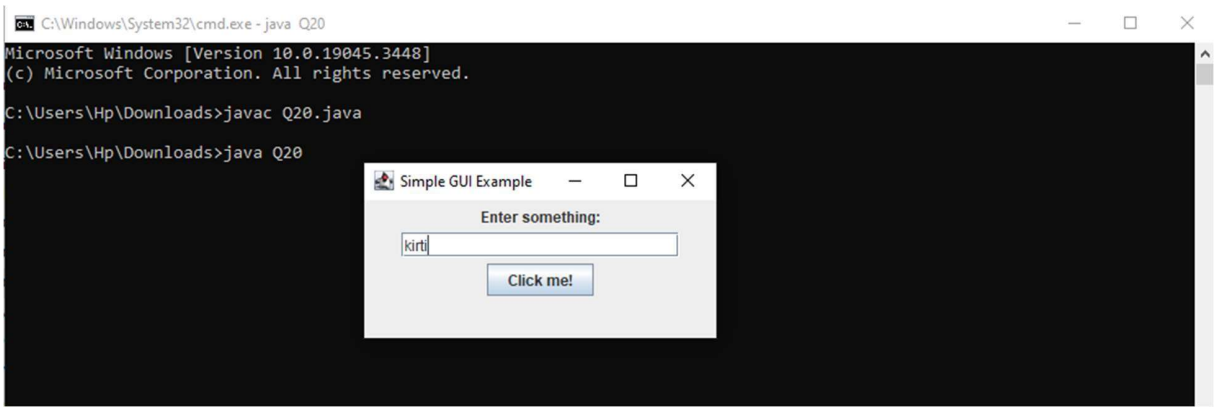
CODE:

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

public class Q20 {
   public static void main(String[] args) {
      JFrame frame = new JFrame("Simple GUI Example");
      frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
      frame.setSize(300, 150);
      frame.setLayout(new FlowLayout());
      JLabel label = new JLabel("Enter something:");
      frame.add(label);
      JTextField textField = new JTextField(20);
      frame.add(textField);
      JButton button = new JButton("Click me!");
      frame.add(button);
      button.addActionListener(new ActionListener() {
         @Override
         public void actionPerformed(ActionEvent e) {
            String userText = textField.getText();
            JOptionPane.showMessageDialog(frame, "You entered: " + userText);
         }
      });
      frame.setVisible(true);
   }
}
```


```
import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

public class Q20 {
   public static void main(String[] args) {
      JFrame frame = new JFrame("Simple GUI Example");
      frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
      frame.setSize(300, 150);
```

```
        frame.setLayout(new FlowLayout());
        JLabel label = new JLabel("Enter something:");
        frame.add(label);
        JTextField textField = new JTextField(20);
        frame.add(textField);
        JButton button = new JButton("Click me!");
        frame.add(button);
        button.addActionListener(new ActionListener() {
            @Override
            public void actionPerformed(ActionEvent e) {
                String userText = textField.getText();
                JOptionPane.showMessageDialog(frame, "You entered: " + userText);
            }
        });
        frame.setVisible(true);
    }
}
```

Output:



Program-21

Aim:

WAP to show different layouts using AWT controls.

Code:

```
import java.awt.*;
import java.awt.event.*;
public class Q21 {
   private Frame frame;
   public Q21() {
      frame = new Frame("Layout Demo");
      Button flowButton = new Button("FlowLayout");
      Button borderButton = new Button("BorderLayout");
      Button gridButton = new Button("GridLayout");
      flowButton.addActionListener(new ActionListener() {
         public void actionPerformed(ActionEvent e) {
            setFlowLayout();
         }
      });
      borderButton.addActionListener(new ActionListener() {
         public void actionPerformed(ActionEvent e) {
            setBorderLayout();
         }
      });
      gridButton.addActionListener(new ActionListener() {
         public void actionPerformed(ActionEvent e) {
            setGridLayout();
         }
      });
      Panel buttonPanel = new Panel();
      buttonPanel.add(flowButton);
      buttonPanel.add(borderButton);
      buttonPanel.add(gridButton);
      frame.setLayout(new BorderLayout());
      frame.add(buttonPanel, BorderLayout.NORTH);
      frame.setSize(400, 300);
      frame.setVisible(true);
      frame.addWindowListener(new WindowAdapter() {
         public void windowClosing(WindowEvent we) {
            System.exit(0);
         }
      });
   }
   private void setFlowLayout() {
```

```
        frame.setLayout(new FlowLayout());
        frame.validate();
    }
    private void setBorderLayout() {
        frame.setLayout(new BorderLayout());
        frame.validate();
    }
    private void setGridLayout() {
        frame.setLayout(new GridLayout(2, 2));
        frame.validate();
    }
    public static void main(String[] args) {
        new Q21();
    }
}
```

Output:

Program-22

Aim:

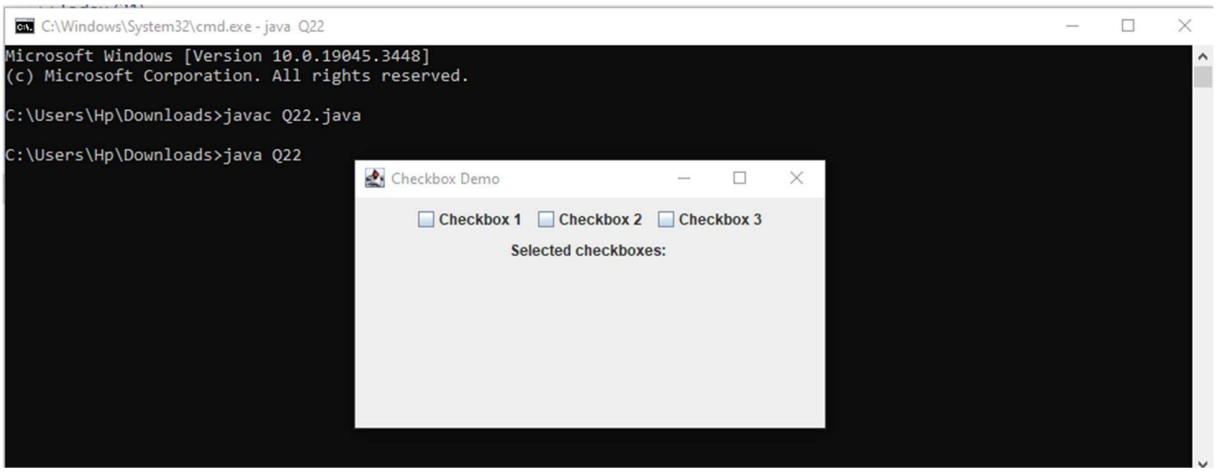
WAP to create a GUI to show checkboxes handling their events.

Code:

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
public class Q22 {
   public static void main(String[] args) {
      JFrame frame = new JFrame("Checkbox Demo");
      frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
      frame.setSize(300, 150);
      frame.setLayout(new FlowLayout());
      JCheckBox checkbox1 = new JCheckBox("Checkbox 1");
      JCheckBox checkbox2 = new JCheckBox("Checkbox 2");
      JCheckBox checkbox3 = new JCheckBox("Checkbox 3");
      JLabel label = new JLabel("Selected checkboxes: ");
      checkbox1.addActionListener(new ActionListener() {
         public void actionPerformed(ActionEvent e) {
            updateLabel(checkbox1, label);
         }
      });
      checkbox2.addActionListener(new ActionListener() {
         public void actionPerformed(ActionEvent e) {
            updateLabel(checkbox2, label);
         }
      });
      checkbox3.addActionListener(new ActionListener() {
         public void actionPerformed(ActionEvent e) {
            updateLabel(checkbox3, label);
         }
      });
      frame.add(checkbox1);
      frame.add(checkbox2);
      frame.add(checkbox3);
      frame.add(label);
      frame.setVisible(true);
   }
   private static void updateLabel(JCheckBox checkbox, JLabel label) {
      String selectedText = label.getText();
      if (checkbox.isSelected()) {
         selectedText += checkbox.getText() + " ";
```

```
        } else {
            selectedText = selectedText.replace(checkbox.getText() + " ", "");
        }
        label.setText(selectedText);
    }
}
```

Output:



Program-23

Aim:

WAP to show the use of Console class for reading and writing.

Code:

```
import java.io.Console;
public class Q23 {
   public static void main(String[] args) {
      Console console = System.console();

      if (console == null) {
         System.out.println("Console is not available. Please run this program in a
console/terminal.");
         System.exit(1);
      }
      console.printf("Welcome to the Console Demo\n");
      String name = console.readLine("Enter your name: ");
      char[] password = console.readPassword("Enter your password: ");
      console.printf("Hello, %s! Your password is: %s%n", name, new String(password));
      for (int i = 0; i < password.length; i++) {
         password[i] = ' ';
      }
      console.format("Password has been cleared from memory.%n");
      console.writer().println("This is a sample output to the console.");
   }
}
```

Output:

Program-24

Aim:

WAP to count the number of characters, words and lines in a file.

Code:

```
import java.io.*;
public class Q24 {
   public static void main(String[] args) {
      String filePath = "sample.txt";
      int charCount = 0;
      int wordCount = 0;
      int lineCount = 0;
      try (BufferedReader reader = new BufferedReader(new FileReader(filePath))) {
         String line;
         while ((line = reader.readLine()) != null) {
            lineCount++;
            charCount += line.length();
            String[] words = line.split("\\s+");
            wordCount += words.length;
         }
      } catch (IOException e) {
         e.printStackTrace();
      }
      System.out.println("Character count: " + charCount);
      System.out.println("Word count: " + wordCount);
      System.out.println("Line count: " + lineCount);
   }
}
```

Output:

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.3448]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Hp\Downloads>javac Q24.java

C:\Users\Hp\Downloads>java Q24
Character count: 648
Word count: 102
Line count: 9

C:\Users\Hp\Downloads>
```